Network Science

Lecture 03: Strong and Weak Ties

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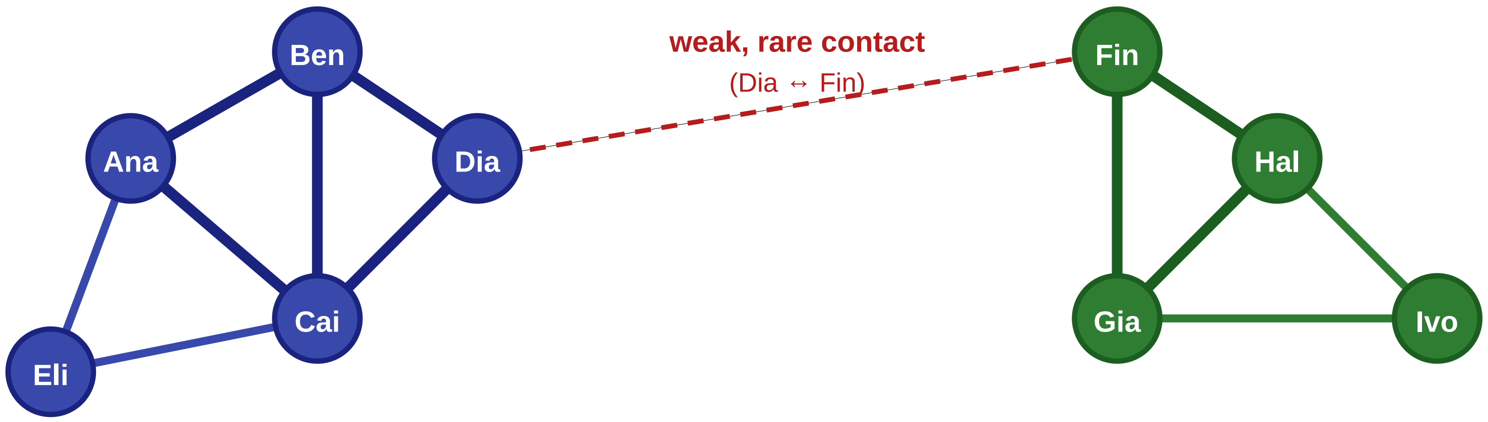
From Graphs to Social Structure

Last time we built the formal language of graphs. Today we use it to ask a different kind of question: once we have the graph, how do we explain what happens on it?

A Motivating Scenario

A small workplace: five questions in one picture

nodes = employees · thick edges = close contacts · thin edges = acquaintances



Team A — dense core

Team B — dense core

Questions we'd like to answer from this picture

① Will Eli and Ben become friends? ② How do we separate close from distant ties?
③ Which single edge controls all information flow between teams? ④ Where would a rumor spread fastest? ⑤ Which tie would we cut to isolate a leak?

> Two teams, each tightly connected internally, linked only by a rare contact between Dia and Fin. Thick lines are close daily contacts, the thin dashed line is an occasional one.

The Catch: Theory vs. Measurement

Each of those five questions has a **theoretical** form and a **measurement** form — and they are not the same.

- The theoretical form asks: *what structure would explain this behavior if we could see everything?*
- The measurement form asks: *what can we actually compute from a real graph where tie strengths, intentions, and labels are hidden?*

Today's lecture lives in the gap between those two.

Today

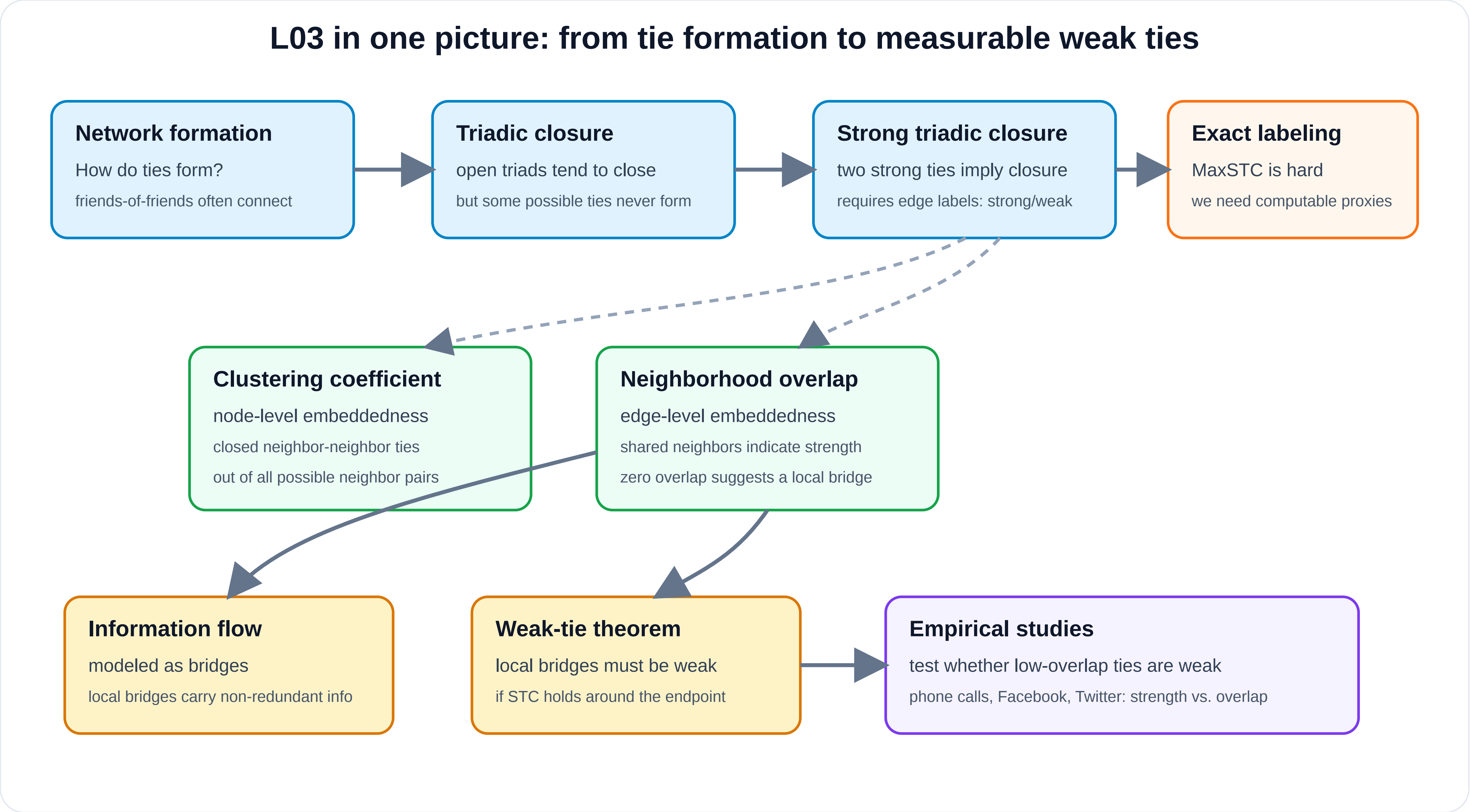
- triadic closure as a theory of friendship formation
- typed edges and Strong Triadic Closure as an edge-labeling model
- why the exact labeling is NP-hard — and what to do instead
- structural heuristics: clustering coefficient and neighborhood overlap
- the weak-tie theorem and its empirical test at scale

In a bit more detail

Every question below has a **theoretical** form and a **measurement** form. The lecture deliberately treats them as distinct tasks.

Question	Theoretical construct	Measurable proxy
How do new ties form?	Triadic closure	Triangle density over time
Are ties of different kinds?	Strong / weak edge labeling + STC	Clustering coefficient
Which edges carry novel info?	Local bridge	Neighborhood overlap
Why are bridges weak?	Weak-Tie Theorem	Tie-strength vs. overlap plots
Which ties keep the network together?	Robustness under edge removal	Giant-component size

One Argument, Many Measures



Takeaway

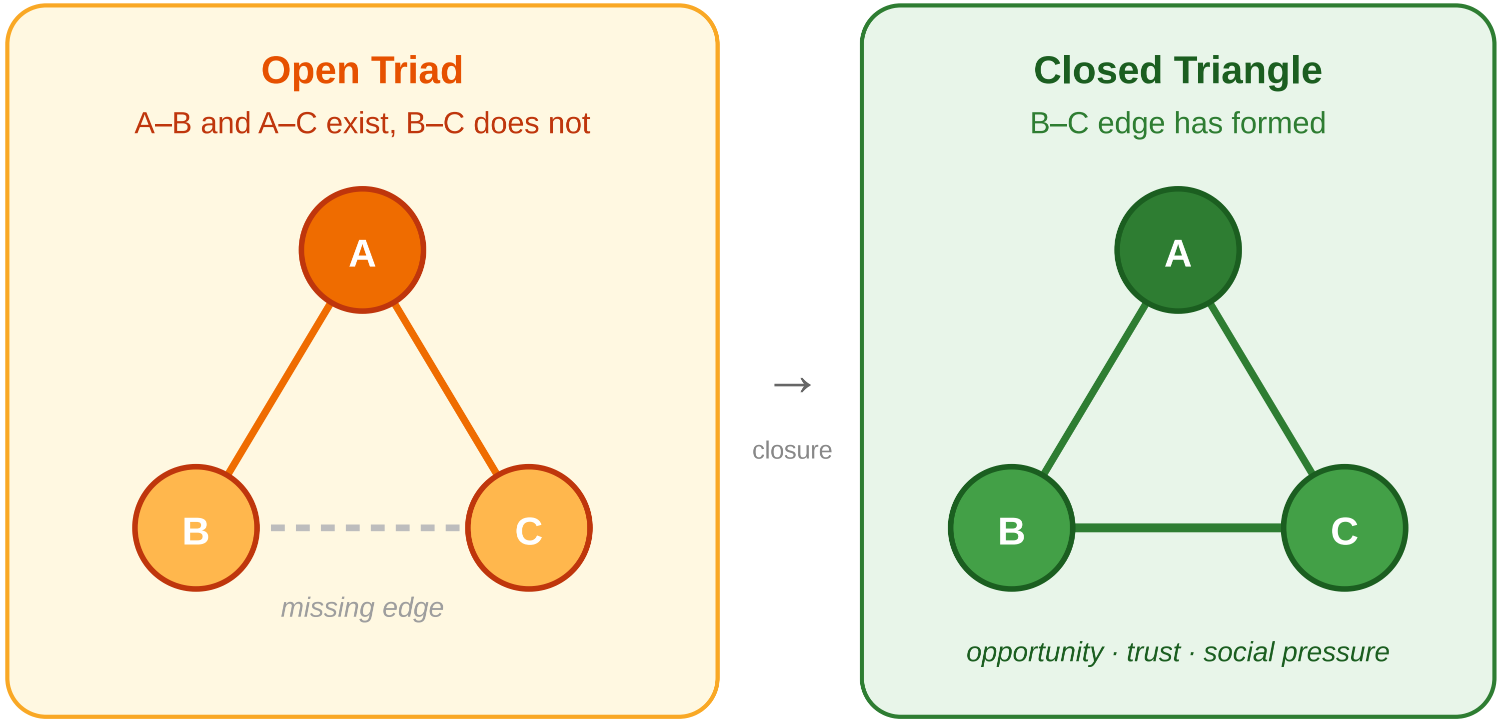
Network science in this lecture is a loop: we build a theoretical model of edge formation and tie strength, discover that exact inference from data is intractable, replace the model with polynomial-time structural proxies, and test whether the proxies match a theoretical prediction at scale.

Triadic Closure

Guiding Question: Why do friends-of-friends so often become linked — and what does that imply for how social graphs grow?

Open Triads and Closure

Triadic Closure. If a node A has edges to two distinct nodes B and C, the probability that an edge (B, C) forms over time is significantly higher than for two random nodes.



The dashed edge in the open triad is the missing one. Triadic closure predicts it will tend to appear over time, producing the closed triangle on the right.

Triadic Closure as a Model

Triadic closure is not an observation — it is a claim about how edges appear. As a model, it makes predictions:

- the graph should accumulate triangles over time
- average distances should shrink
- nodes with many neighbors should have increasingly clustered neighborhoods
- long-range ties should become rarer relative to local ones

Empirical Test: University Email Network

> **Kossinets & Watts (2006),** *Empirical Analysis of an Evolving Social Network*, **Science 311**, 88–90.
>
> They reconstructed the social network of **43 553 students, faculty, and staff** at a large US university from one year of time-stamped email headers, tracking which pairs started exchanging messages over time.

Headline result. Two people who shared **one** mutual email contact were roughly **30 times** more likely to begin emailing each other than two people with no mutual contact. The effect grew monotonically with the number of shared contacts — direct, longitudinal evidence for triadic closure at scale.

What the Study Could Not See

Kossinets & Watts observed only one signal per edge: *did an email pass between these two people, yes or no?*

- They could not see whether an edge was a daily collaboration or a one-off administrative ping.
- They could not distinguish a pair who spoke for hours off-platform from a pair who had never met.
- Every edge in the dataset was, by construction, **binary and untyped**.

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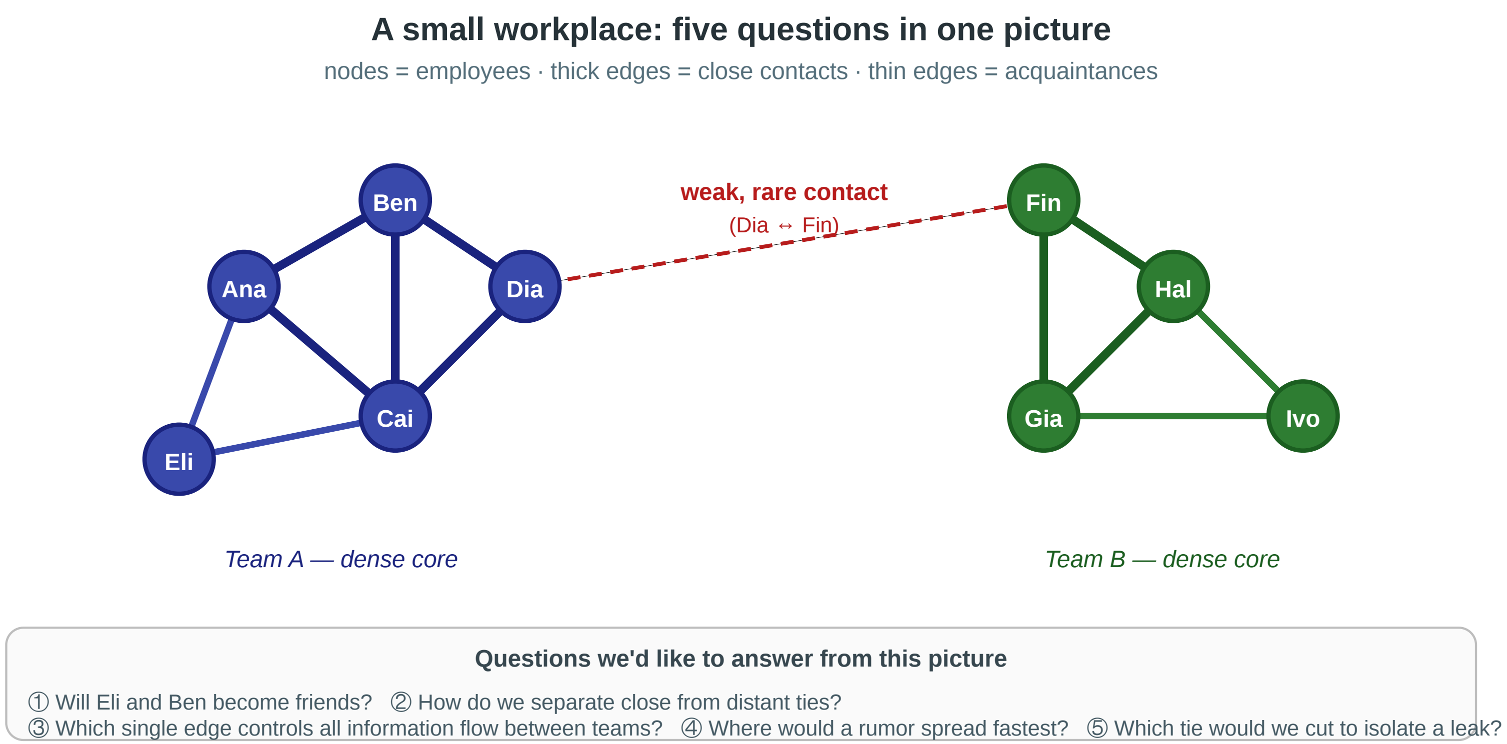
Closure is visible in their data. **Tie strength is not.** The next module takes that limitation seriously: if we want to model strong vs. weak ties, we need a formal object — an edge labeling — that the email graph never gave us.

Takeaway

Triadic closure is a generative theory of edge formation. Kossinets & Watts confirmed it at scale on a university email network — but their binary email data could not distinguish strong from weak ties. To go further, we need to model tie strength explicitly.

Quick Check



Assume triadic closure is the mechanism shaping this workplace network.

1. Which **missing edge** in the picture would triadic closure most naturally predict to appear next?
2. If that edge does **not** appear, does this already refute the triadic-closure model? Why or why not?

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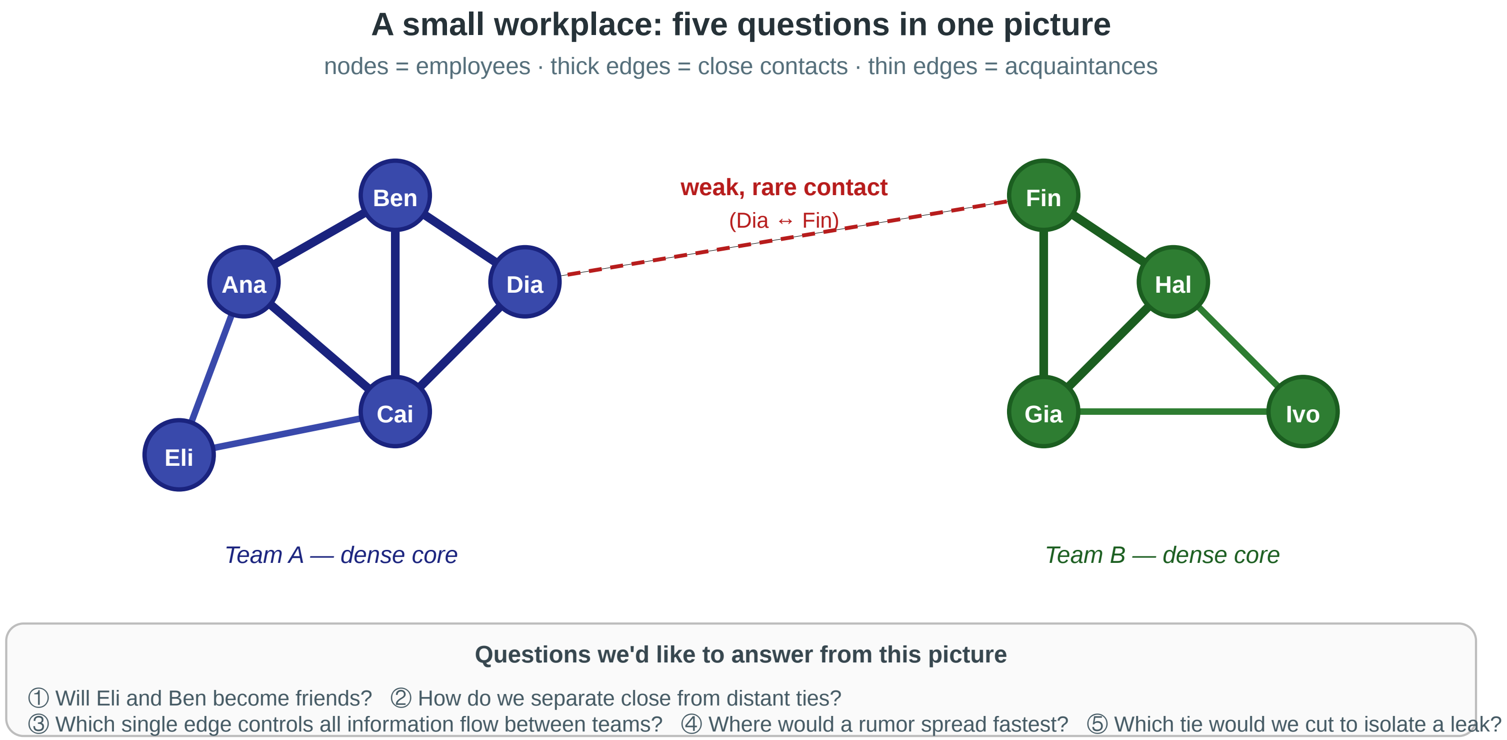

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Solution

1. Triadic closure predicts that the most likely new edge is the one that would **close an open triad**: two people who already share a common contact but are not yet directly connected.

2. A single failed prediction does not refute a probabilistic model. But *persistent* failure across many such triples would indicate either that the closure mechanism is weak in this population or that some competing mechanism — interpersonal conflict,

Typed Edges and Strong Triadic Closure

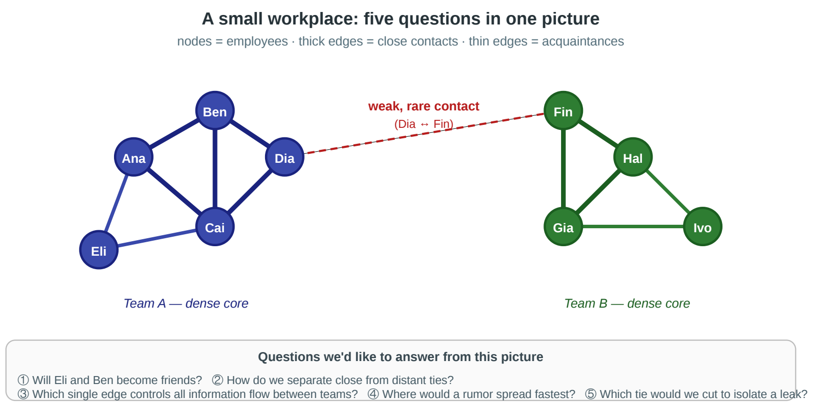
Guiding Question: If ties come in different *kinds*, what constraint does that place on how they coexist around a node?

Edges Are Not All the Same

In the scenario, the Ana–Ben edge and the Dia–Fin edge are clearly different:

- Ana–Ben is a daily interaction between people in the same team
- Dia–Fin is a rare, occasional contact between people in different teams

Graph theory's default edge is binary: either present or absent. We now extend it.



Edge labeling. A graph $G = (V, E)$ together with a function

$$\ell : E \to \mathrm{Strong}, \mathrm{Weak}$$

assigning a *type* to every edge. We write $S(v)$ for the set of strong neighbors of v under ℓ.

Strong Triadic Closure as a Constraint on Labelings

Strong Triadic Closure (STC). A labeling ℓ of G satisfies STC if, for every node v and every pair $u_1, u_2 \in S(v)$ of strong neighbors of v, the edge (u_1, u_2) exists in E — *with any label* (Strong or Weak). STC constrains which pairs must be **connected**, not how the closing edge is labeled.

The Recovery Problem — MaxSTC

In real data we observe the graph G but not the labeling ℓ. Given G, we want to recover the labeling that best explains the observed structure — the one with as many strong edges as possible subject to STC.

> **MaxSTC.** Given $G = (V, E)$, find a labeling ℓ satisfying STC that maximizes $|e \in E : \ell(e) = \mathrm{Strong}|$.

Equivalently, the decision version asks: given G and an integer k, is there an STC-satisfying labeling with at least k strong edges?

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Complexity of MaxSTC

> **Result (Sintos & Tsaparas, 2014).** MaxSTC is **NP-hard** as an optimization problem and **NP-complete** as a decision problem on general graphs. Polynomial-time algorithms exist for certain restricted graph classes (e.g., cographs and several bipartite-like structures).

Consequence: on any realistically sized social network, we **cannot** recover the true STC-maximizing labeling exactly.

This is a **modeling problem**, not just an engineering problem. The theoretical construct we care about is computationally out of reach. We need a different strategy.

What We Do Instead

Since we cannot recover the true labeling, we take a different route:

1. Define structural **heuristics** that approximate the "strong"/"weak" character of edges in polynomial time.
2. Derive a theoretical **prediction** from STC (the weak-tie theorem).
3. **Test** the prediction on real data, using the heuristics as proxies for the unobservable labeling.

This is the plan for the next three modules.

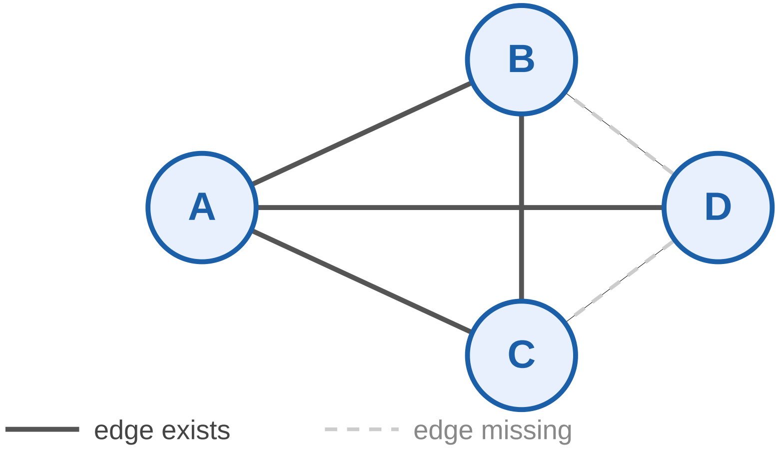
Takeaway

Tie strength is most precisely modeled as an edge labeling, and Strong Triadic Closure is a constraint on such labelings. Recovering the best labeling is NP-hard — so we must approximate the exact theory with polynomial-time structural heuristics.

Quick Check

Consider four nodes A, B, C, D with edges A–B, A–C, A–D, and B–C. No other edges exist (dashed = missing).

1. Find a labeling of this graph that satisfies STC with the maximum number of strong edges.
2. Is the optimal labeling unique?
3. How many possible labellings are there



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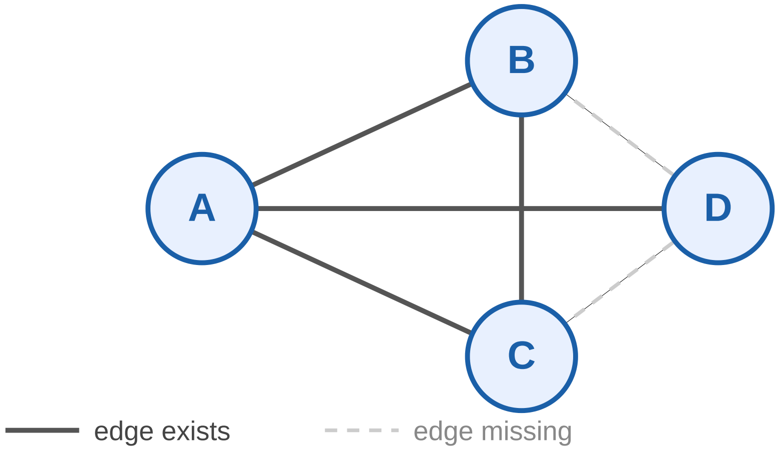


Solution

1 +3. Optimal labeling. We have $2^4 = 16$ possible labellings. At node A, making all three ties strong would force B–D, C–D, and B–C to exist. Only B–C exists, so at most *two* of A's ties can be strong — and they must be the two whose endpoints are actually connected: A–B and A–C. The edge B–C can be strong without further constraint. So:

$$\ell(A{-}B) = \ell(A{-}C) = \ell(B{-}C) = S, \quad \ell(A{-}D) = W$$

gives **three** strong edges — the maximum.

2. Uniqueness. In this small graph, yes. On larger graphs, MaxSTC can have exponentially many optimal labelings — another reason the problem is hard.

Structural Heuristics for Tie Strength

Guiding Question: Without access to the true edge labels, how can we estimate which parts of a network behave like "strong, embedded" regions and which behave like "weak, bridging" ones?

Clustering Coefficient

Local clustering coefficient. For a node i with degree k_i:

$$C_i = \frac{\text{edges among } i\text{'s neighbors}}{\binom{k_i}{2}}$$

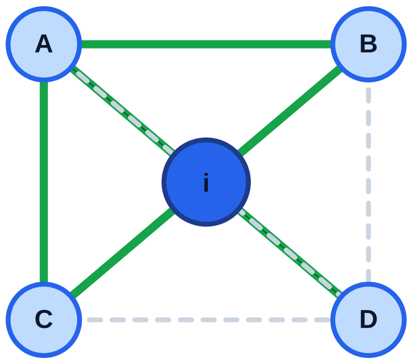
$C_i \in [0,1]$.

- $C_i = 1$: all possible neighbor-neighbor ties exist
- $C_i = 0$: no neighbor-neighbor ties exist

Heuristic link to STC. If v has many strong ties, those endpoints must be mutually connected. A high C_v therefore signals a neighborhood that *could* be labeled mostly strong.

Local clustering coefficient: existing friend ties out of all possible friend ties

One focal node with k = 4 neighbors



solid green = existing neighbor-neighbor tie

dashed gray = possible but missing

Why the denominator is binomial

Among k neighbors, each possible friend relation is an unordered pair.

A	B	C	D

All possible pairs: **AB, AC, AD, BC, BD, CD**

number of possible pairs
C(4,2) = 4·3 / 2 = 6

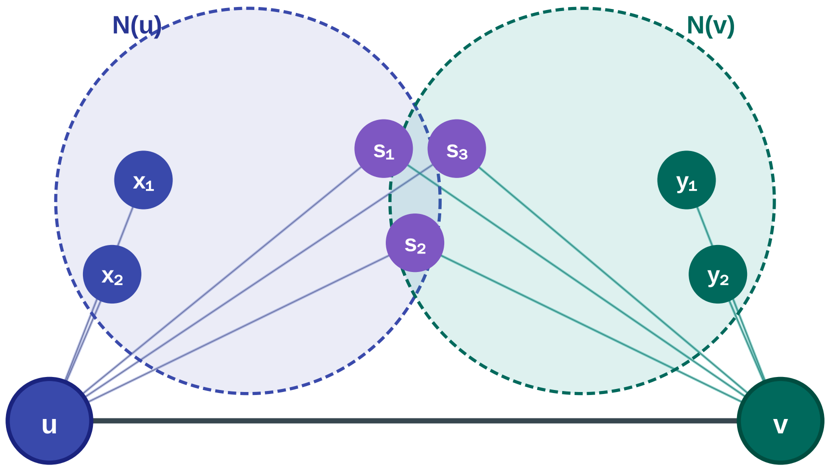
Here: 3 existing out of 6 possible **C_i = 3/6 = 0.5**

Neighborhood Overlap

Neighborhood overlap. For an edge (u, v):

$$O(u,v) = \frac{|N(u) \cap N(v)|}{|N(u) \cup N(v)|}$$

where $N(u)$ and $N(v)$ exclude each other.



- $O = 0$ — pure local bridge: no shared neighbors
- $O = 1$ — edge inside a complete clique
- Jaccard similarity of the two neighbor sets

Why These Heuristics Instead of Exact Labels

	Exact MaxSTC	Clustering coeff.	Overlap
Scope	whole graph	per node	per edge
Complexity	NP-hard	$O(k^2)$ per node	$O(k)$ per edge
Captures	true tie type	neighborhood density	edge embeddedness
Feasible on real graphs?	No	Yes	Yes

Here k is the **degree** of the node (or of an edge's endpoint) — i.e. its number of neighbors. Clustering coefficient checks all $\binom{k}{2}$ pairs of a node's neighbors; overlap intersects the two endpoints' neighbor sets.

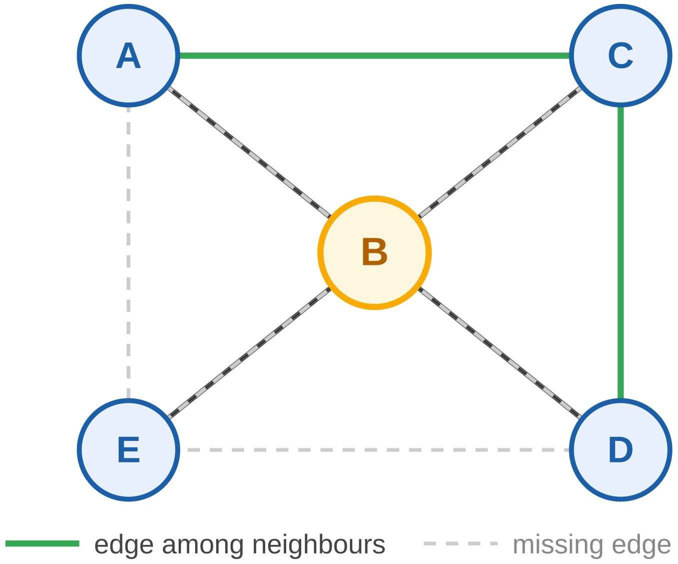
Takeaway

Clustering coefficient and neighborhood overlap exist because MaxSTC is intractable. They are polynomial-time structural proxies for a theoretical object we cannot compute — trading precision for feasibility on real-world networks.

Quick Check

Node B has 4 neighbors: A, C, D, E. Among B's neighbors, exactly 2 edges exist (green): (A, C) and (C, D).

1. Compute C_B.
2. Under STC, what is the maximum number of strong ties B can have?
3. What does C_B tell you about B's structural role that STC alone does not?



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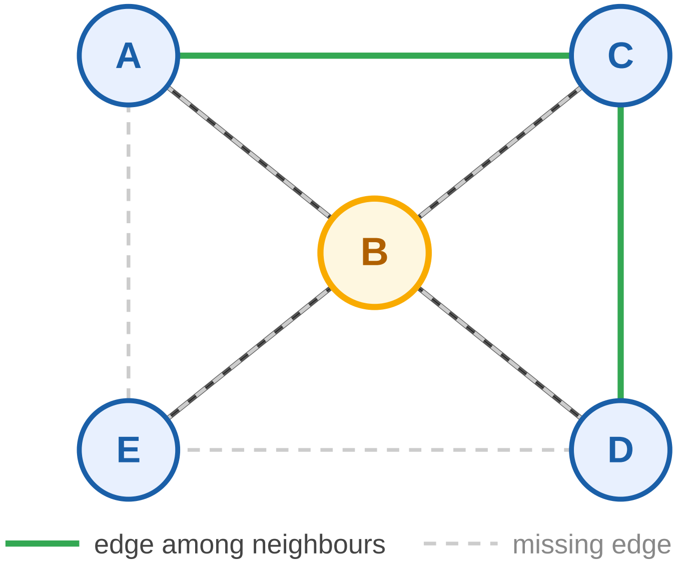


Quick Check — Solution

1. $\binom{4}{2} = 6$ possible edges, 2 actual $\Rightarrow C_B = 1/3$.

2. Maximum strong ties from B is **2**. The only pairs of B's neighbors connected by an edge are (A, C) and (C, D); a third strong tie would force a currently missing edge (e.g. A–D), violating STC.

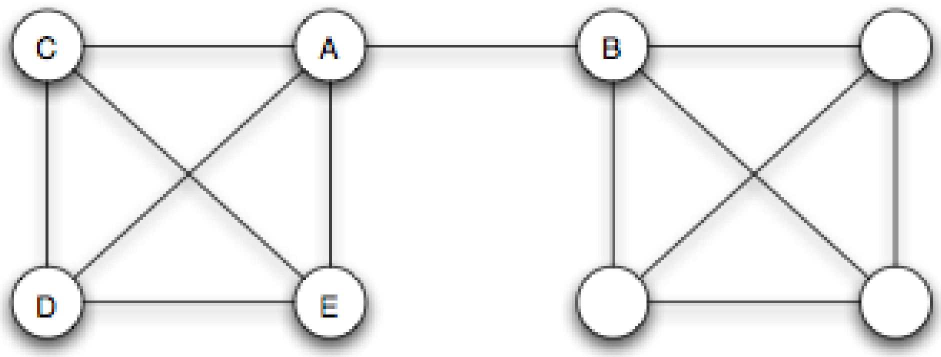
3. The clustering coefficient is a *continuous* structural measurement — $C_B = 1/3$ tells us B sits in a moderately sparse neighborhood, suggesting a potential brokerage position between the sub-clusters A, C and D. STC by itself gives only a hard upper bound on the number of strong ties but no information about the *shape* of the neighborhood. The heuristic sacrifices exactness for expressive power.

Bridges, Local Bridges, and the Weak-Tie Theorem

Guiding Question: Which edges are structurally most important for keeping information flowing across a network — and can we predict from theory which ones those will be?

Bridge

Bridge. An edge $e \in E$ is a *bridge* if removing it increases the number of connected components. Equivalently: e lies on no cycle.

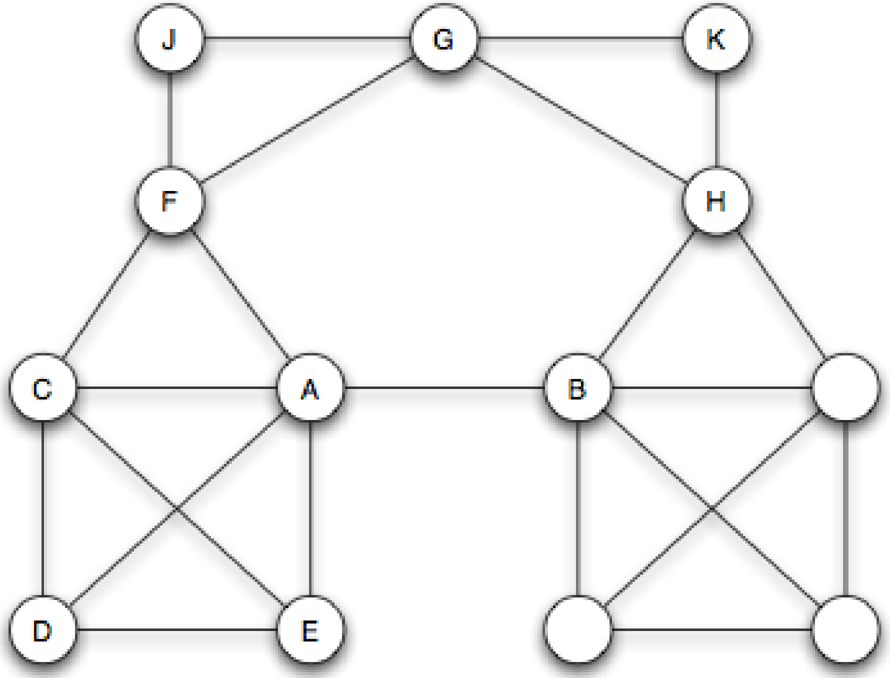


Source: Easley & Kleinberg 2010

Removing the highlighted edge splits the graph in two. The endpoints' distance becomes infinite — they can no longer reach each other through any other route.

Local Bridge

Local bridge. An edge (u, v) is a *local bridge* if u and v have no neighbors in common: $N(u) \cap N(v) = \emptyset$. Equivalently: $O(u, v) = 0$.



Source: Easley & Kleinberg 2010

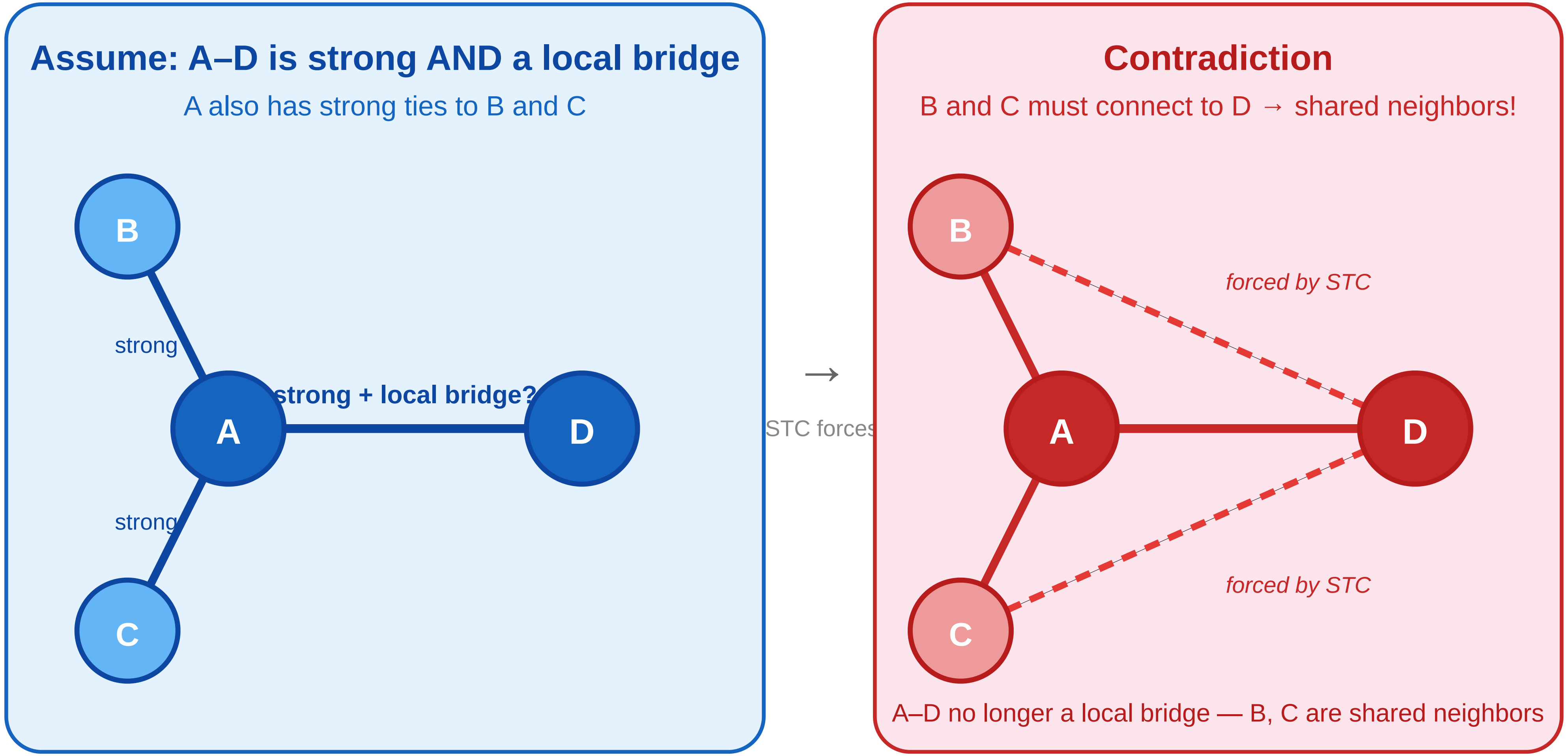
A local bridge does not need to disconnect the graph — its endpoints may still reach each other via a long detour. Note that *local bridge* is already expressible in terms of the neighborhood-overlap heuristic: an edge with $O = 0$ is exactly a local bridge.

The Weak-Tie Theorem

Theorem (Granovetter). If a node a satisfies Strong Triadic Closure and is incident to at least two strong ties, then any local bridge incident to a must be labeled Weak.

This is the theoretical prediction we will test. It links the labeling model (STC, tie types) to a measurable structural property (local bridges) via a local if–then statement.

Proof Sketch by Contradiction



Assume A–D is both **strong** and a **local bridge**. Since A has another strong tie, say A–B, STC forces B–D to exist. But then B is a shared neighbor of A and D — destroying the local-bridge condition.

Contradiction.

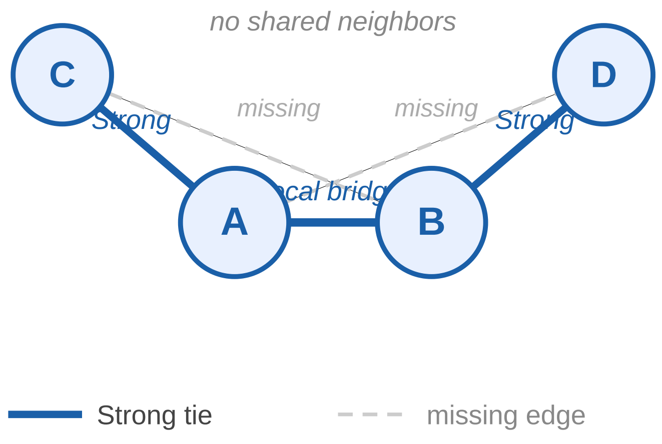
Takeaway

For nodes satisfying STC with at least two strong ties, incident local bridges must be weak ties. This is the theoretical prediction that connects our unobservable labeling model to the measurable neighborhood-overlap heuristic — and it is the prediction we now test against real data.

Quick Check

A–C and B–D are labeled **Strong**. The edge A–B is the only link between the two sides; dashed edges are missing.

1. Is A–B a local bridge?
2. Could A–B also be labeled **Strong** under STC?



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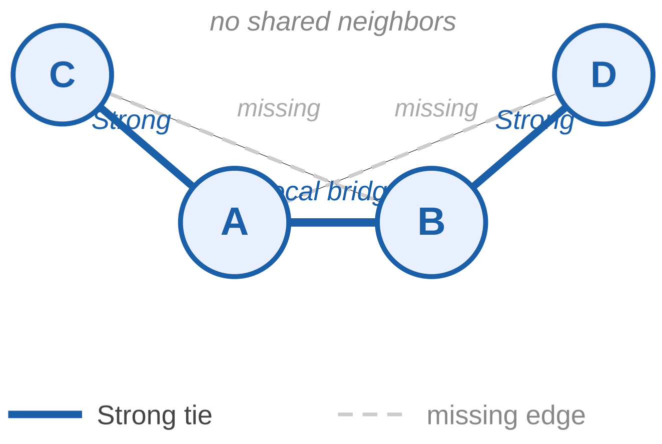


Quick Check — Solution

1. Yes. $N(A) = B, C$ and $N(B) = A, D$, so A and B have no common neighbor.

2. No. If A–B were Strong, then STC at A would require B–C, and STC at B would require A–D. Both are missing. Therefore, any STC-satisfying labeling that keeps A–C and B–D strong must label the local bridge A–B as **Weak**.

Empirical Validation at Scale

Guiding Question: Does the theoretical prediction — low overlap correlates with low tie strength — survive contact with real-world data at scale?

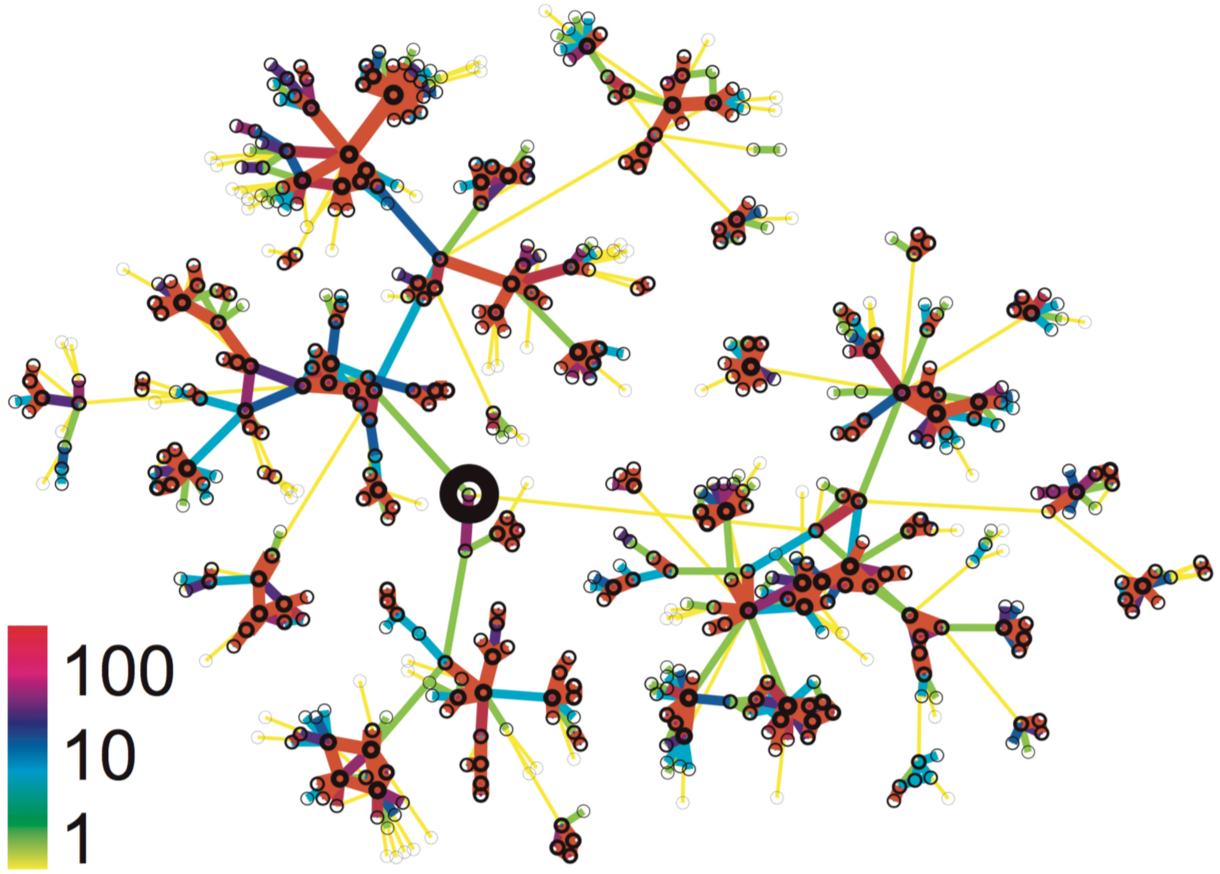
Study 1: Cell-Phone Network (Onnela et al. 2007)

Source: Onnela et al. 2007

Question	Does tie strength correlate with structural embeddedness in a real large-scale network?
Data	Nationwide mobile call logs — 18 months, ~7 M users, one operator
Graph	Undirected weighted: nodes = phone numbers, edge weight = cumulative call duration
Measures	Tie-strength proxy: call duration (percentile); structural proxy: neighborhood overlap $O(u, v)$; global: giant-component size
Prediction	Weak-tie theorem: low-overlap edges are weak, high-overlap edges are strong — monotone relationship

Limitations: Call duration misses topic, reciprocity, and face-to-face contact. Single country and operator limits

Local ego-network of one subscriber. Dense cluster of high-duration contacts; peripheral nodes are reached only by low-duration, low-overlap ties.



generalizability. No ground-truth labels to validate the proxy.

Source: Onnela et al. 2007

Study 1 — Result 1: Overlap vs. Tie Strength

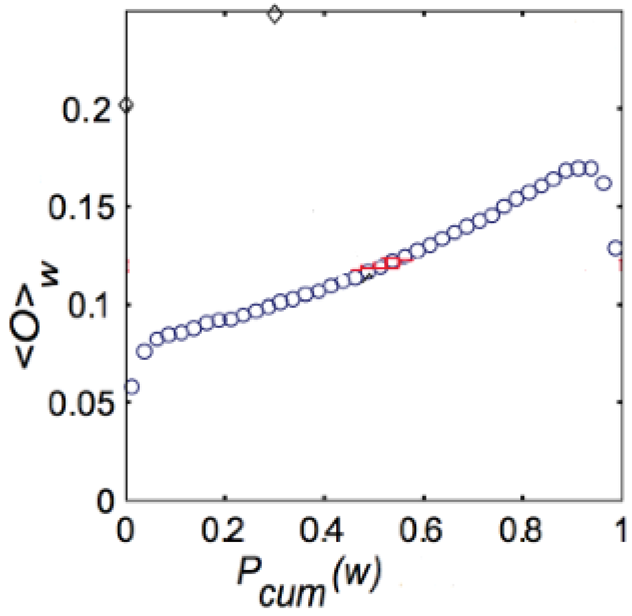
Source: Onnela et al. 2007

Question	Do low-overlap edges tend to be weak and high-overlap edges strong?
Procedure	Bin all edges by call-duration percentile; compute mean $O(u, v)$ per bin
X-axis	Tie-strength percentile (weakest left → strongest right)
Y-axis	Mean neighborhood overlap per bin

$\langle O \rangle$ is the **average neighborhood overlap** of edges in a tie-strength bin. $P_{\text{cum}}(w)$ is the **cumulative tie-strength percentile**: values near 0 are the weakest call-duration ties, values near 1 are the strongest.

Outcome: Clear, smooth monotone relationship — weakest ties cluster near $O \approx 0$; strongest ties accumulate at high O.

Interpretation: Two independently computable proxies align exactly as the weak-tie theorem predicts — not a threshold but a continuous gradient.



Source: Onnela et al. 2007

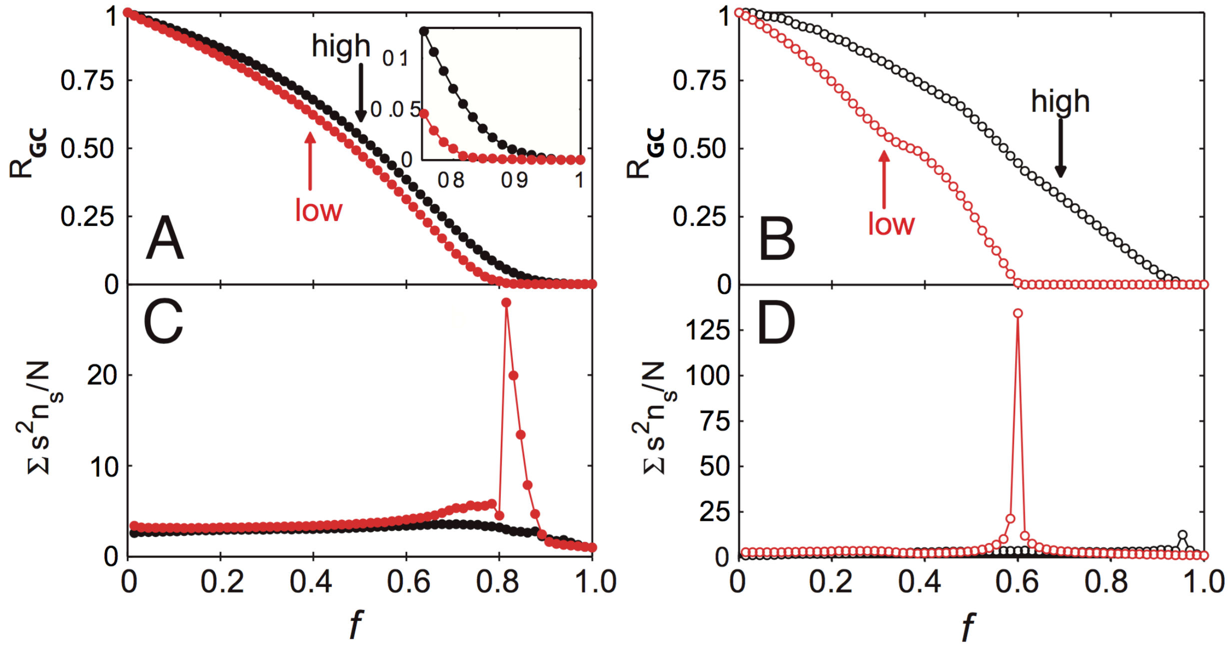
Limitations: Both axes are proxies. Geographic proximity could independently inflate both call duration and overlap, confounding the tie-type interpretation.

Study 1 — Result 2: Knockout Experiment

Source: Onnela et al. 2007

Question	Are weak ties structurally load-bearing or peripheral?
Procedure	Remove edges in two orders: weakest-first (low call duration) vs. strongest-first
Measure	Giant connected component size after each removal step
Condition A	Weakest ties removed first
Condition B	Strongest ties removed first (control)

Axis notation: f is the fraction of removed edges. R_{GC} is the fraction of all nodes still in the giant connected component. $\sum_s s^2 n_s / N$ summarizes fragmentation outside the giant component: n_s is the number of components of size s, and peaks mark the point where the network breaks apart.



Source: Onnela et al. 2007

Outcome: Weakest-first removal fragments the giant component far faster than strongest-first — weak ties are the network's connective tissue.

Interpretation: Confirms the structural role the theorem assigns: local bridges (weak ties) hold the global network together.

Limitations: Removal is a static simulation, not a real intervention. Giant-component size is coarse — betweenness or average path length might show finer differences. Removal order uses call duration as proxy, not true tie labels.

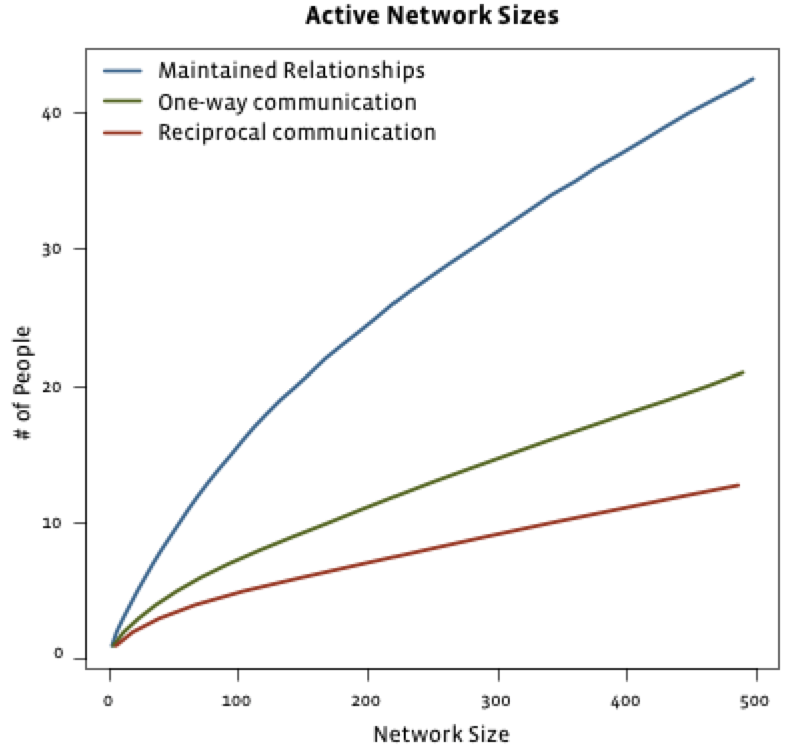
Study 2: Facebook — Maintained vs. Total Friendships

Source: Gon\ccalves et al. 2011

Question	Do total online friends and actively maintained ties scale the same way?
Data	Facebook friendship and interaction logs across users with varying friend counts
Measure A	Total declared friends (any structural link)
Measure B	Maintained friends (reciprocal messages / comments / wall posts)

Outcome: Total friends grow without bound. Maintained ties plateau around ~150 — consistent with Dunbar's cognitive limit on strong-tie capacity.

Interpretation: Platforms remove the cost of weak ties but leave the cognitive budget for strong ties unchanged. The weak/strong distinction is amplified, not collapsed.



Source: Easley & Kleinberg 2010

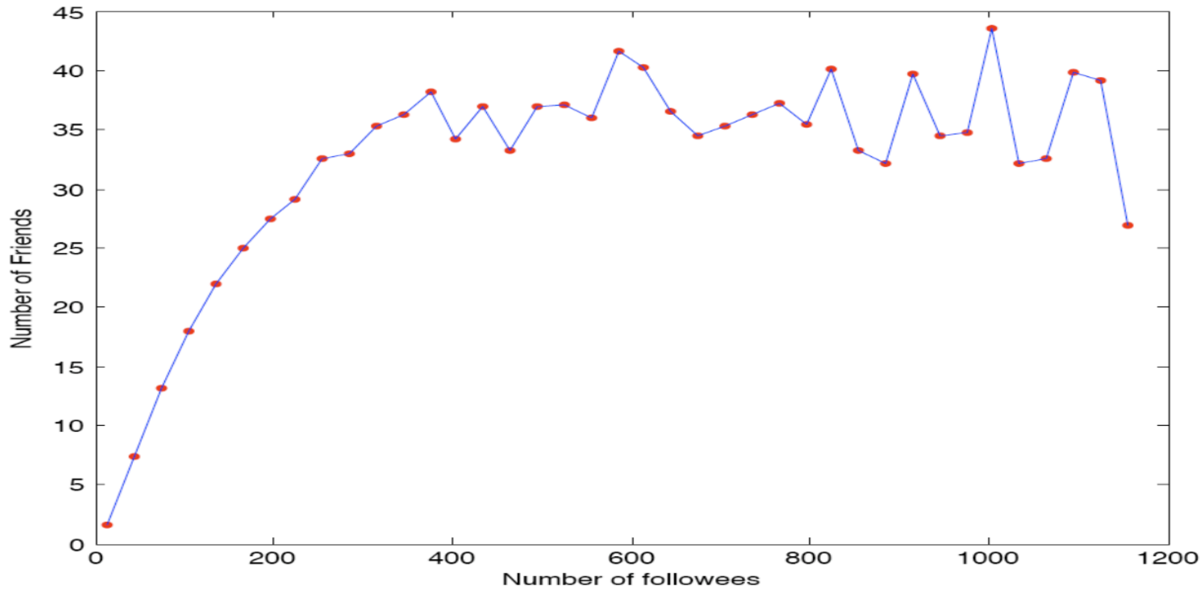
Limitations: "Maintained" captures only on-platform interactions; offline contact is invisible. The plateau may partly reflect News Feed ranking rather than purely cognitive limits. Dunbar's ~150 number is itself debated.

Study 3: Twitter — Followers vs. Interacting Friends

Source: Huberman et al. 2008

Question	Does follower count predict active interaction count on an asymmetric platform?
Study	Huberman, Romero & Wu, arXiv:0812.1045, 2008: "Social networks that matter: Twitter under the microscope"
Data	Twitter activity logs
Measure A	Followers: any account that subscribed (passive, asymmetric)
Measure B	Active friends: accounts the user sent ≥ 2 directed messages to

Outcome: Follower count scales freely (power-law). Active interacting friends plateau at a small number, independent of follower count. **Interpretation:** Same pattern as Study 2, different platform. Following is cheap; active communication is cognitively bounded. Weak ties scale with affordances; strong ties do not.



Source: Huberman et al. 2008

Limitations: Twitter following is often parasocial — it does not map cleanly onto "weak tie." The ≥ 2-message threshold is arbitrary. Algorithmic timelines shape interaction independently of genuine tie strength.

Study 4: Facebook Information Diffusion

Source: Bakshy et al. 2012

Question	Who spreads web links on Facebook: strong ties or weak ties?
Study	Bakshy, Rosenn, Marlow & Adamic, WWW 2012
Design	Field experiment randomizing exposure to friends' URL-sharing signals in News Feed
Scale	253 million subjects
Tie strength	Four prior-interaction counts: messages, comments, photo co-tags, shared comment threads
Weak tie cutoff	Aggregate comparison: $k = 0$ prior interactions = weak; $k \geq 1$ = strong

This is a direct online test of Granovetter's idea: tie strength is measured, exposure is experimentally varied, and the outcome is actual sharing behavior.

Finding: Strong ties are more influential *per exposure*, but weak ties generate most information diffusion overall because they are far more numerous.

Novelty: Weak-tie exposure has the largest relative effect because it surfaces links users were less likely to discover independently.

Interpretation: Strong ties transmit well locally; weak ties carry diverse information across the broader network.

Limitations: Tie strength is inferred from Facebook interaction logs, not self-reported relationship quality. The platform's News Feed ranking is part of the treatment environment.

Closing the Loop

We set out to test a theoretical model (typed edges + STC) on data where the true labels are unobservable and the exact recovery problem is NP-hard. The strategy was:

1. Define polynomial-time structural heuristics (clustering coefficient, overlap)
2. Derive a prediction from the theory (weak-tie theorem)
3. Measure the heuristics on real data and compare

The Onnela and social-media results close the loop: the heuristics behave exactly as the theorem predicts, and the same structural logic explains why weak ties carry novel information online. The model earns its keep even though we could never compute its exact solution.

Takeaway

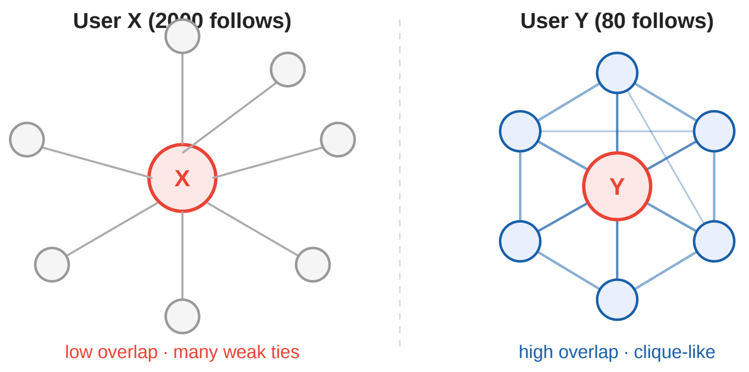
The theoretical model (typed edges, STC, weak-tie theorem) cannot be inferred exactly from data, but its prediction — low overlap edges are weak — is confirmed at scale by independent empirical proxies. Model and measurement close the loop.

Quick Check

X follows 2000 accounts; most are unconnected to each other.

Y follows 80 accounts; most follow each other back.

1. Using neighborhood overlap as a proxy, whose connections look more "strong-tie-like"?
2. What does the weak-tie theorem predict about the diversity of information each student receives?



Quick Check — Solution

1. Y's connections look more **strong-tie-like**: dense mutual follow cluster → high average neighborhood overlap. X's large following includes many algorithm-driven follows with near-zero overlap → low average, bridge-like.

2. The weak-tie theorem predicts:

- **X** receives diverse, non-redundant information — structural signature of many weak, low-overlap edges acting as local bridges.
- **Y** receives redundant, reinforced information — structural signature of a dense clique with few outgoing shortcuts.

The prediction matches the everyday pattern of algorithm-curated feeds vs. closed-friend echo chambers.

Wrap-Up

- **Modeling** gave us triadic closure, typed edges, STC, and the weak-tie theorem
- **Measurement** forced us to abandon the exact labeling (NP-hard) and adopt polynomial-time proxies: clustering coefficient and neighborhood overlap
- **The theorem** provided the prediction that connects the two
- **Empirical data** at scale (Onnela, Facebook, Twitter) confirm the prediction

Reading

- Chapter 3 of Easley & Kleinberg (2010).
- Kossinets & Watts, *Empirical Analysis of an Evolving Social Network*, Science **311**, 88–90 (2006) — longitudinal evidence for triadic closure on a university email network.
- Sintos & Tsaparas, *Using Strong Triadic Closure to Characterize Ties in Social Networks*, KDD 2014 — NP-hardness of MaxSTC and tractable special cases.

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Image Credits & References

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